# International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 9

# Roadmap

1. The Fisher effect
2. PPP and Reality
3. PPP and income comparison
4. Price levels and income
   - The Balassa-Samuelson effect
     (and the Bhagwati-Kravis-Lipsey effect)

# Absolute and Relative PPP (from last time)

- Absolute PPP is what we so far called PPP

$$E_{\$|€} = P_{US}/P_E$$

- Relative PPP (weaker than absolute PPP)

$$\frac{E_{\$|€,t} - E_{\$|€,t-1}}{E_{\$|€,t-1}} = \pi_{US,t} - \pi_{E,t}$$

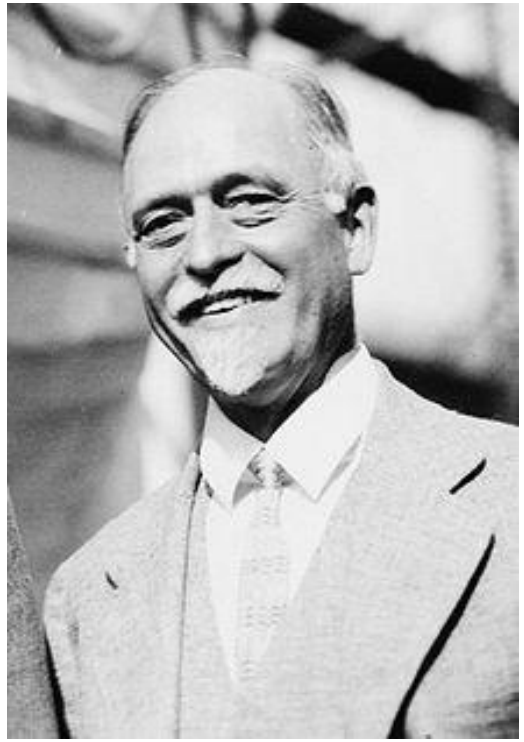
  – $\pi_{US,t} = (P_{US,t} - P_{US,t-1})/P_{US,t-1}$ is inflation in the US, $\pi_{E,t}$ - in Eurozone

- Issue
  – Absolute PPP needs identical baskets of goods in both countries to hold
  – Relative PPP likely overcomes the different baskets issue

# The Fisher Effect

# Irving Fisher (1867-1947)

- Contribution to economics
  - The interest parity, the Fisher effect, the theory of debt deflation, etc.
  - "the greatest economist the United States has ever produced“ (Milton Friedman)
- Predictions
  - Oct 15, 1929: “Stock prices have reached what looks like a permanently high plateau”
  - Oct 28-29, 1929: Dow Jones Industrial average falls by 25%



# The Fisher Effect

- **UIP**

$$R - R^* = \frac{E^e - E}{E}$$

- Assume that financial markets expect **(relative) PPP** to hold, that is,

$$\frac{E^e - E}{E} = \pi^e - \pi^{*e}$$

- Combine

$$R - R^* = \pi^e - \pi^{*e}$$

- The Fisher effect: *a rise in the domestic inflation rate causes an equal rise in the interest rate on deposits of domestic currency* ***in the long run****, when other factors remain constant*

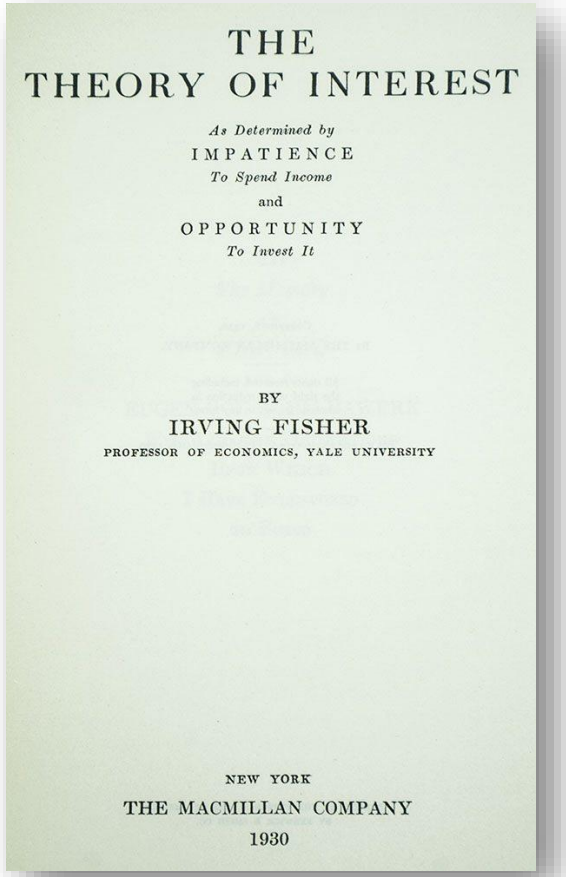
# The Fisher Effect

- UIP + relative PPP

$$R - R^* = \pi^e - \pi^{*e}$$

- Note that

$$R - \pi^e = R^* - \pi^{*e} \Rightarrow \boldsymbol{r = r^*}$$


THE
THEORY OF INTEREST

*As Determined by*
IMPATIENCE
*To Spend Income*
and
OPPORTUNITY
*To Invest It*

BY
IRVING FISHER
PROFESSOR OF ECONOMICS, YALE UNIVERSITY

NEW YORK
THE MACMILLAN COMPANY
1930

# A PPP Model of Long-Run FX

(A PPP Model of FX under Flex Prices)

# A PPP Model of Long-Run FX

$$E_{\$|€} = P_{US}/P_E$$

$$P_{US} = \frac{M_{US}^s}{L(\bar{r}_{US} + \bar{\pi}_{US}, \bar{Y}_{US})}$$

$$P_E = \frac{M_E^s}{L(\bar{r}_E + \bar{\pi}_E, \bar{Y}_E\ )}$$

Predictions

1. $M_{US}^s \uparrow \Rightarrow P_{US} \uparrow, E_{\$|€} \uparrow$
2. $\bar{Y}_{US} \uparrow \Rightarrow P_{US} \downarrow, E_{\$|€} \downarrow$
3. $(\bar{r}_{US} + \bar{\pi}_{US}) \uparrow \Rightarrow P_{US} \uparrow, E_{\$|€} \uparrow$ (in contrast to our short-run analysis )

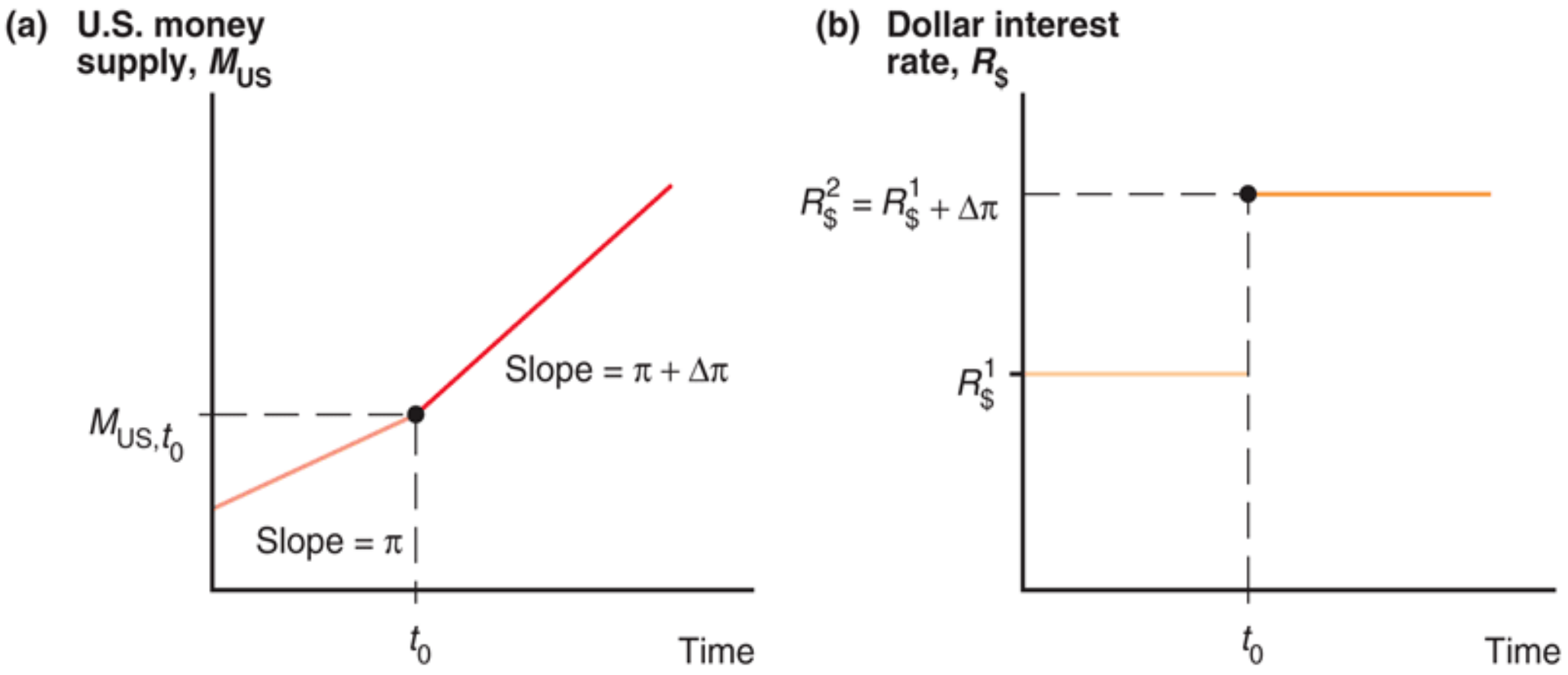
# Exercise

## A permanent increase in money **growth rate**

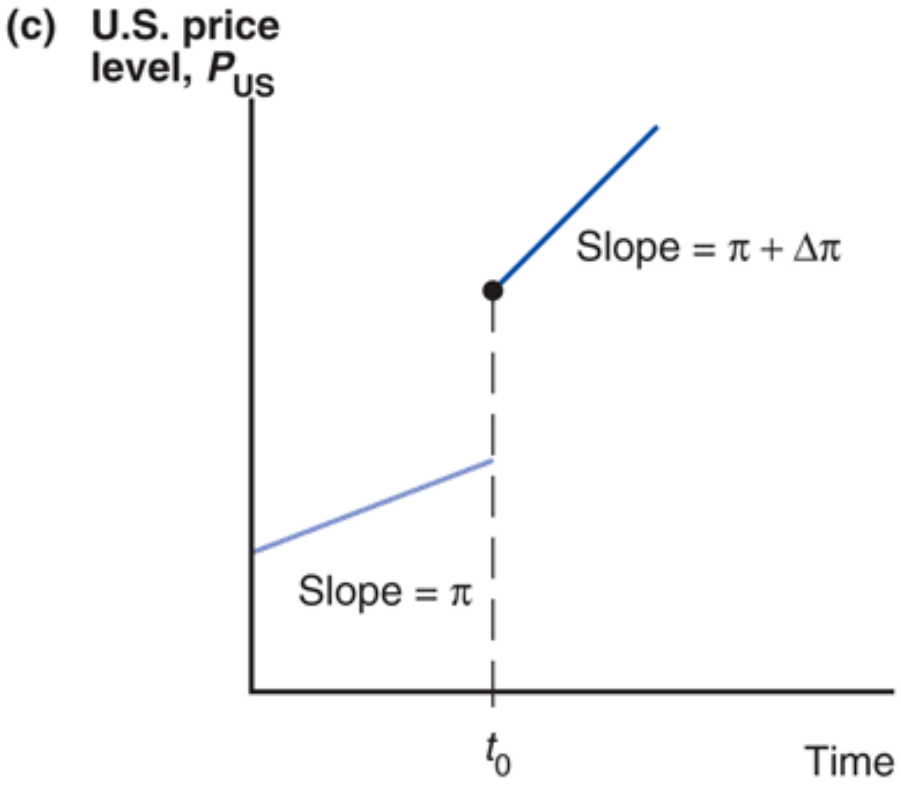
- Home central bank increases the growth rate of the money at time $t_0$ from $\pi$ to $\pi + \Delta\pi$ after $t_0$ (the foreign inflation rate remains the same)
  - For inflation to adjust "fast," we assume that we consider long periods. For example, $t_0$ is 2019*, while* $t_0$+1 is 2022.

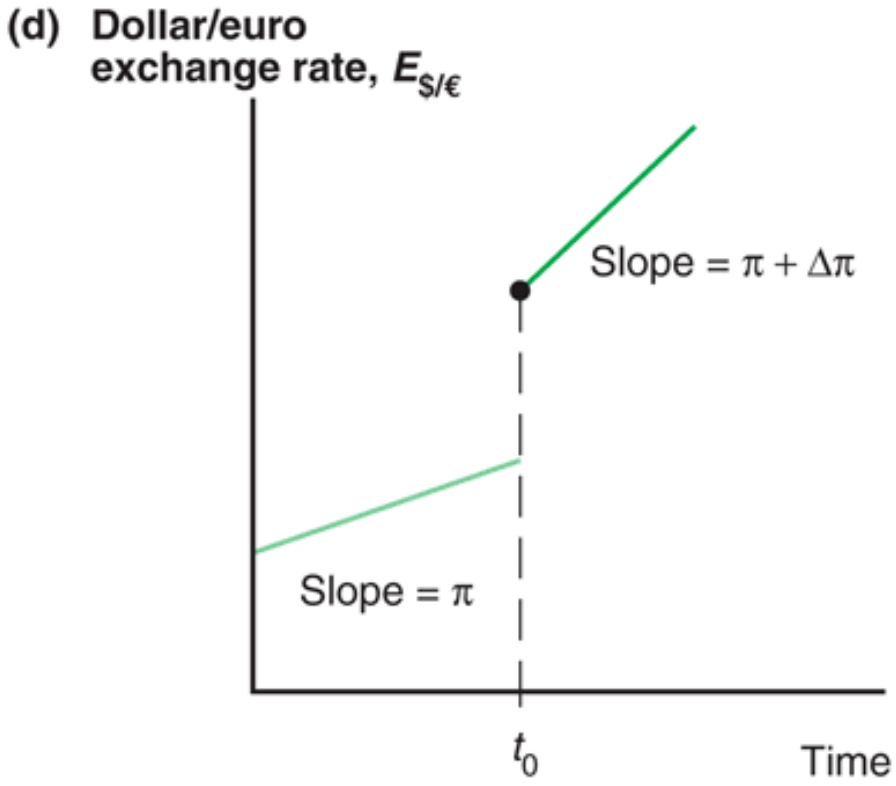
- **According to the Fisher effect**, the domestic interest rate will increase to a new higher level

# A Permanent Increase in Money Growth



# A Permanent Increase in Money Growth




$$P = \frac{M^S}{L(R, Y)}$$

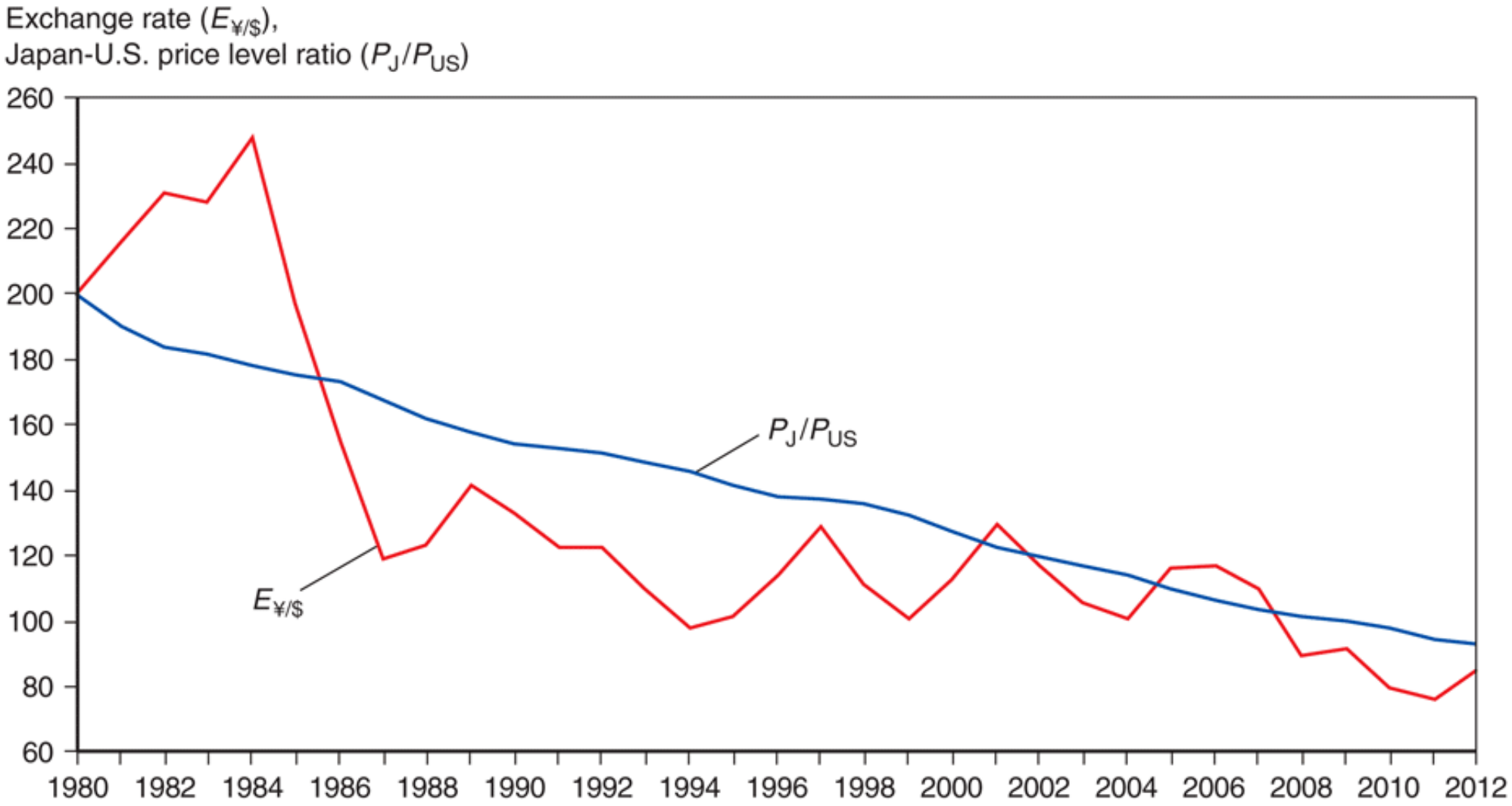
$$E = \frac{P}{P^*}$$

# A Key Takeaway of the Exercise

- To understand how nominal interest rate variations affect the exchange rate, one has to take into account the factors that caused the interest rate change

# PPP and Reality

# JPY/USD and Relative Japan-U.S. Price Levels, 1980–2012

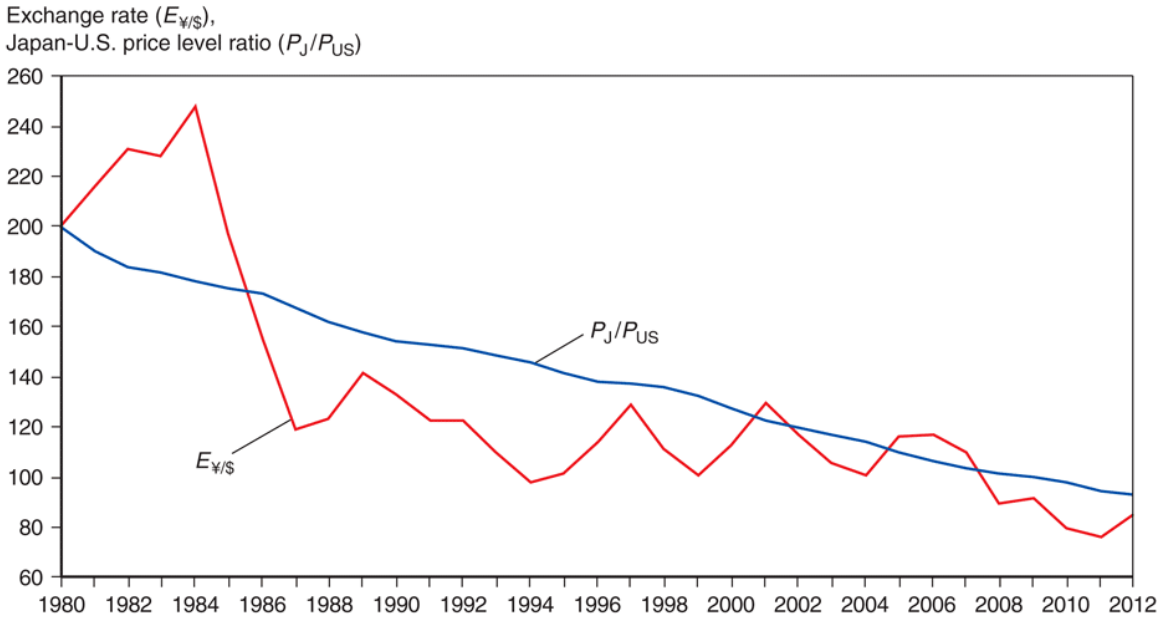



**Source:** IMF, *International Financial Statistics*. Exchange rates and price levels are end-of-year data.

# The Yen/Dollar Exchange Rate and Relative Japan-U.S. Price Levels, 1980–2012

FX and relative price levels do not always move together

- Relative price levels change slowly and have a small range of movement
- FX moves abruptly and experience large fluctuations




**Source:** IMF, *International Financial Statistics*. Exchange rates and price levels are end-of-year data.

But PPP is a long-run theory!

# PPP in the Short Run

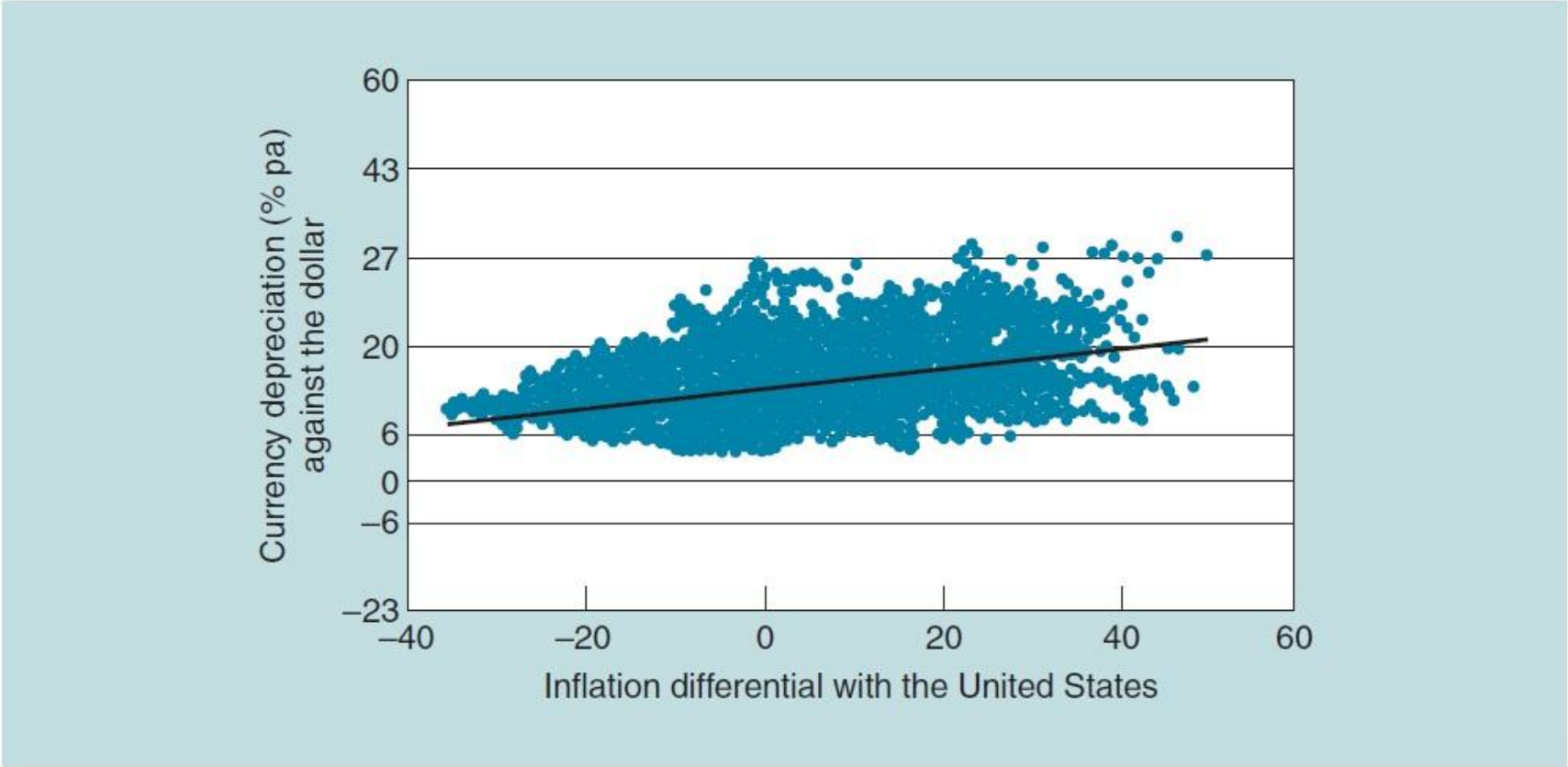
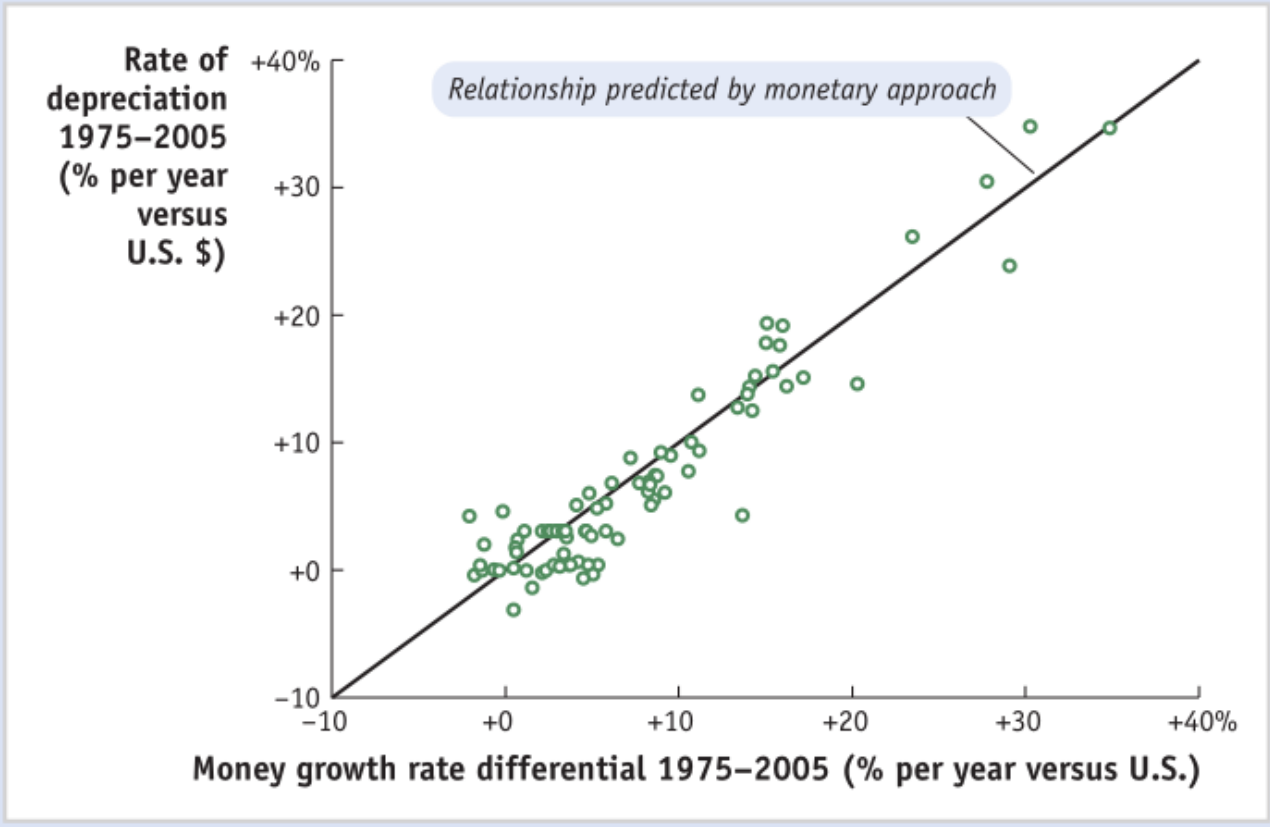



FIGURE 19.5a ● Annual change in exchange rates and inflation, 2010. *Source:* OECD.

# PPP in the Long Run




**Money Growth Rates and the Exchange Rate, 1975–2005** This scatterplot shows the relationship between the rate of exchange rate depreciation against the U.S. dollar and the money growth rate differential versus the United States over the long run, based on data for a sample of 82 countries. The data show a strong correlation between the two variables and a close resemblance to the theoretical prediction of the monetary approach to exchange rates, which would predict that all data points would appear on the 45-degree line.

*Source: IMF, International Financial Statistics.*

The correlation between the two variables is strong and bears a close resemblance to the theoretical prediction of PPP that all data points would appear on the 45-degree line.

# Problems with PPP

# Problems with PPP

1. Trade barriers and non-tradable products

2. Imperfect competition

3. Differences in baskets

# Trade Barriers and Non-tradable Products

- Transport costs and trade restrictions
  - The greater the transport costs, the greater the range over which the exchange rate can deviate from its PPP value
  - Services are often non-traded, that is, very high transportation cost

- LOP empirical support is weak in the data
  - The prices of identical goods, when converted to a single currency, differ substantially across countries

# Example 1 of the LOP Failure

## The Big Mac Index (the Economist, since 1986)

- Focus on a common, specific good

- Locally produced and based on input costs from a wide range of sectors (agriculture, labor, real estate, etc.)

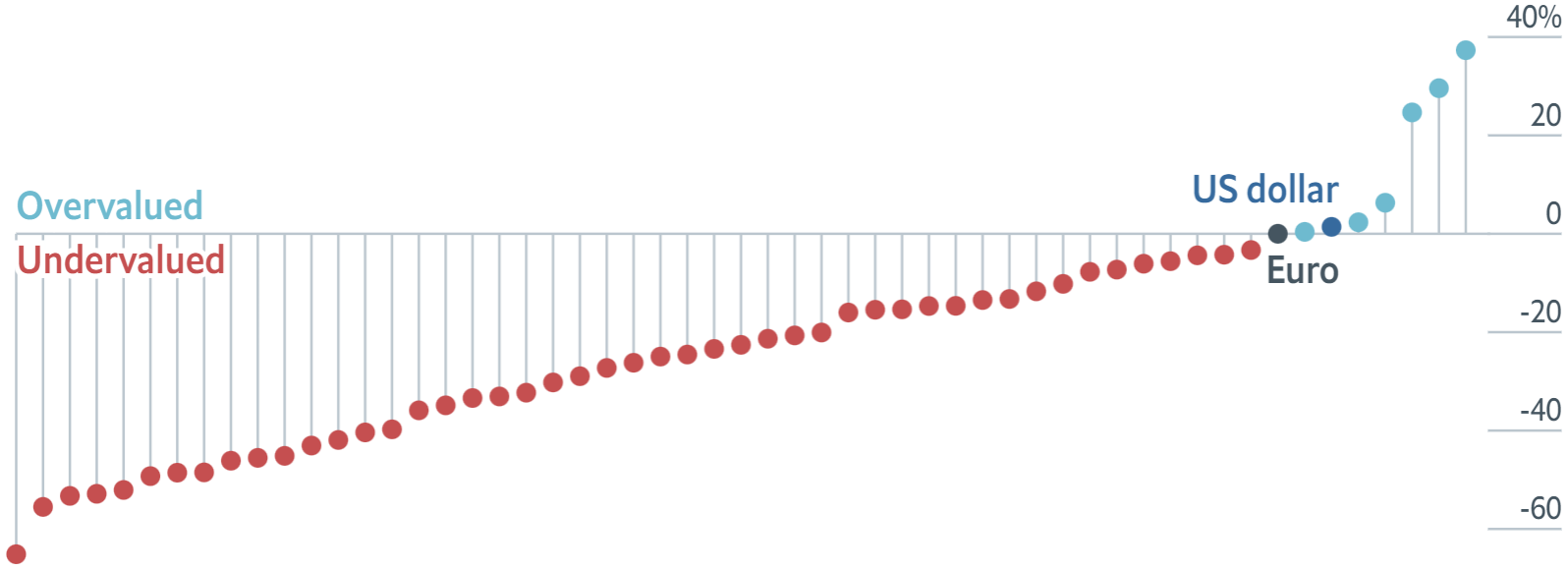
Choose a base currency: Euro

Show index at: Jan 2023

Adjust the index to account for GDP per person: Raw index | GDP-adjusted

## The US dollar is 1.4% overvalued against the euro

January 2023




| Country | | 2000 — 2023 | Under/over valued, % |
|---|---|---|---|
| Switzerland | Franc | | 37.3 |
| Uruguay | Peso | | 29.6 |
| Norway | Krone | | 24.7 |
| Sweden | Krona | | 6.3 |
| Denmark | Krone | | 2.3 |
| United States | US$ | | 1.4 |
| Argentina | Peso | | 0.4 |
| Euro area | Euro | | BASE CURRENCY |
| Australia | A$ | | -3.3 |
| Saudi Arabia | Riyal | | -4.3 |
| Israel | Shekel | | -4.4 |
| Sri Lanka | Rupee | | -5.6 |
| Venezuela | Bolívar | | -48.6 |
| Hong Kong | HK$ | | -49.3 |
| India | Rupee | | -52.1 |
| Ukraine | Hryvnia | | -52.9 |
| Taiwan | NT$ | | -53.3 |
| Indonesia | Rupiah | | -55.5 |
| Egypt | Pound | | -65.1 |

A Big Mac costs US$5.36 in the United States and €4.86 in the euro area. The implied exchange rate is 1.10. The difference between this and the actual exchange rate, 1.09, suggests the US dollar is 1.4% overvalued

# Imperfect Competition

- Imperfect competition + trade frictions or regulations may result in price discrimination (pricing to market)
  - A firm sells the same product for different prices in different markets to maximize profits, based on expectations about what consumers are willing to pay

# The Nominal Exchange Rate

Acquista

€ 1.989,00

Buy

$1,599.00

1 Euro equals

1.04 United States Dollar

**$2069** in Italy

**$1599** in the US

(+ 6-10% sales tax depending on the state)

# Example 2 of the LOP Failure

## The iPhone Index

- Focus on a common, specific good that has identical performance characteristics.

- Unlike Big Mac, iPhone is not produced locally.

- Price differences depend on transportation costs, taxes, and pricing to market rather than on the presence of non-traded, e.g., labor input, as in the case of the Big Mac Index.

## Apple iPhone 11, 64GB, $US

| | | | |
|---|---|---|---|
| Argentina | $1,964.44 | Greece | $915.48 |
| Vietnam | $1,250.04 | Israel | $910.12 |
| Czech Republic | $1,247.33 | Romania | $902.20 |
| Serbia | $1,110.33 | NZ | $899.33 |
| Sweden | $1,049.66 | Latvia | $891.67 |
| South Africa | $1,045.97 | Switzerland | $888.85 |
| Denmark | $1,003.03 | **Australia** | **$875.18** |
| Italy | $998.81 | Saudi Arabia | $874.40 |
| Philippines | $988.47 | India | $870.85 |
| Finland | $986.90 | Slovakia | $866.67 |
| Portugal | $986.90 | Indonesia | $866.40 |
| Ireland | $970.83 | Taiwan | $849.83 |
| Spain | $963.10 | Singapore | $838.69 |
| France | $963.10 | Canada (Ontario) | $838.08 |
| Netherlands | $963.10 | South Korea | $833.34 |
| Belgium | $963.10 | Mexico | $822.32 |
| Hungary | $961.19 | Malaysia | $819.04 |
| Austria | $951.19 | China | $805.12 |
| Slovenia | $951.19 | UAE | $803.54 |
| Croatia | $941.76 | Russia | $800.93 |
| Norway | $940.20 | Thailand | $796.29 |
| Brazil | $939.66 | Japan tax | $775.06 |
| Bulgaria | $938.18 | Hong Kong | $774.06 |
| UK | $934.62 | US (Los Angeles) | $765.41 |
| Luxembourg | $931.26 | *Canada ex-tax* | $741.67 |
| Germany | $927.20 | *Japan ex-tax* | $704.60 |
| Macedonia | $922.04 | *US ex-tax* | $699.00 |

Source: CommSec, Sep, 2020

# Differences in Baskets

- Levels of average prices (*P*) differ across countries because of differences in how representative baskets of goods and services are measured
  - Ex: Italy – more coffee, Argentina – more steaks, etc.

- Relative PPP partly addresses this issue
  - However, relative price changes can lead to deviations from relative PPP

# PPP and Income Comparison

# PPP and Income Comparison

- **US in 2018**
  - GDP = $20.5 trillion
  - Population = 326,766,748
  - GDP per capita = $62,700

- **China in 2018**
  - GDP = 91.2 trillion yuan
  - Population  = 1,415,045,928
  - GDP per capita = 64,500 yuan
  - GDP per capita at market FX = $9,700 (1/6.5 of the US level)

# How to compare income levels across countries?

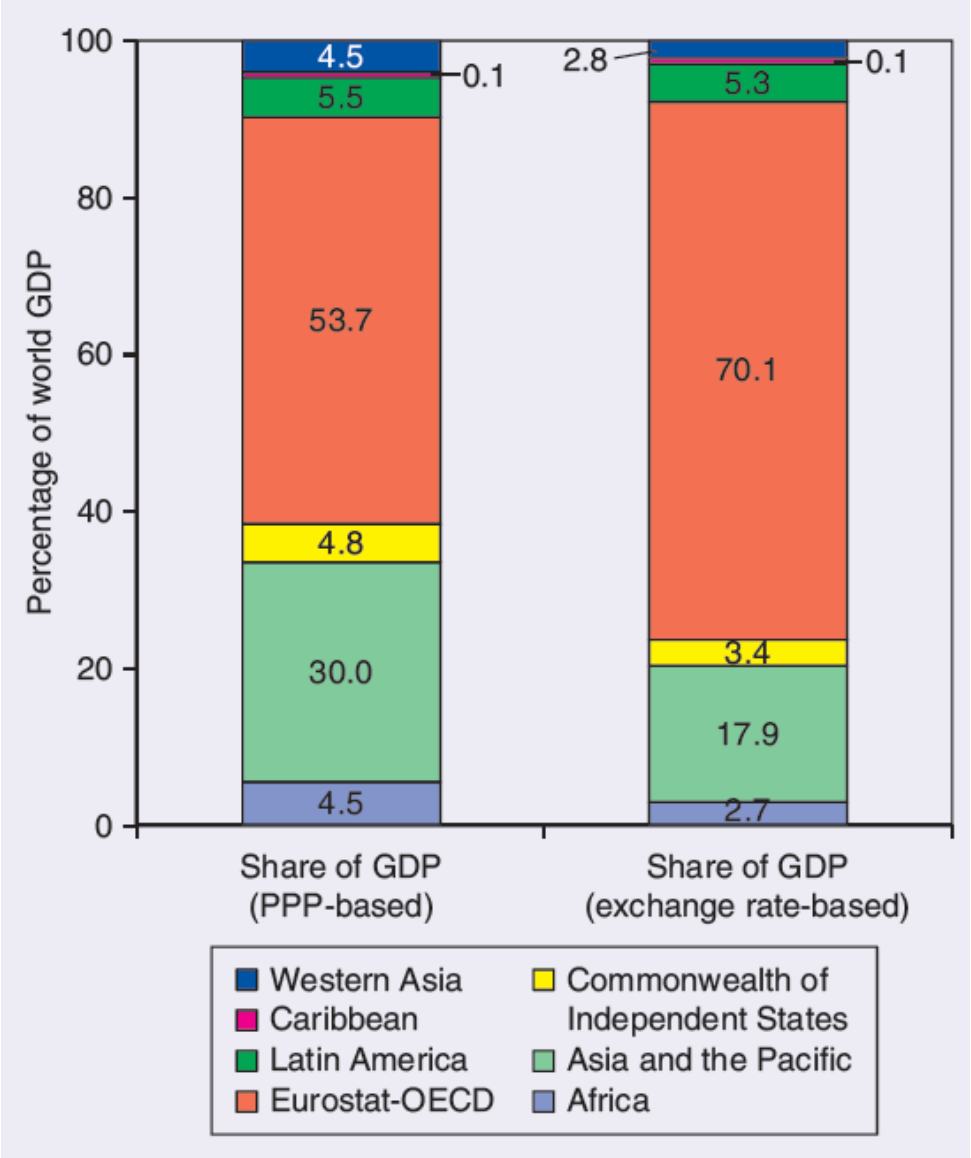
- There are two issues
  1. Conversion of GDP using the market FX contaminates the comparison with the factors that influence the FX in the short run.
  2. Due to the failures of the LOP, same goods can have different prices in different countries. As a result, identical incomes buy different quantities in different countries
- Solution: use the PPP exchange rate
  - It equalizes the value of comparable market baskets of goods and services between two countries

# How to compare income levels across countries?

- China GDP per capita in 2018 at PPP
  = $17,000 (x2 its value at the market FX)

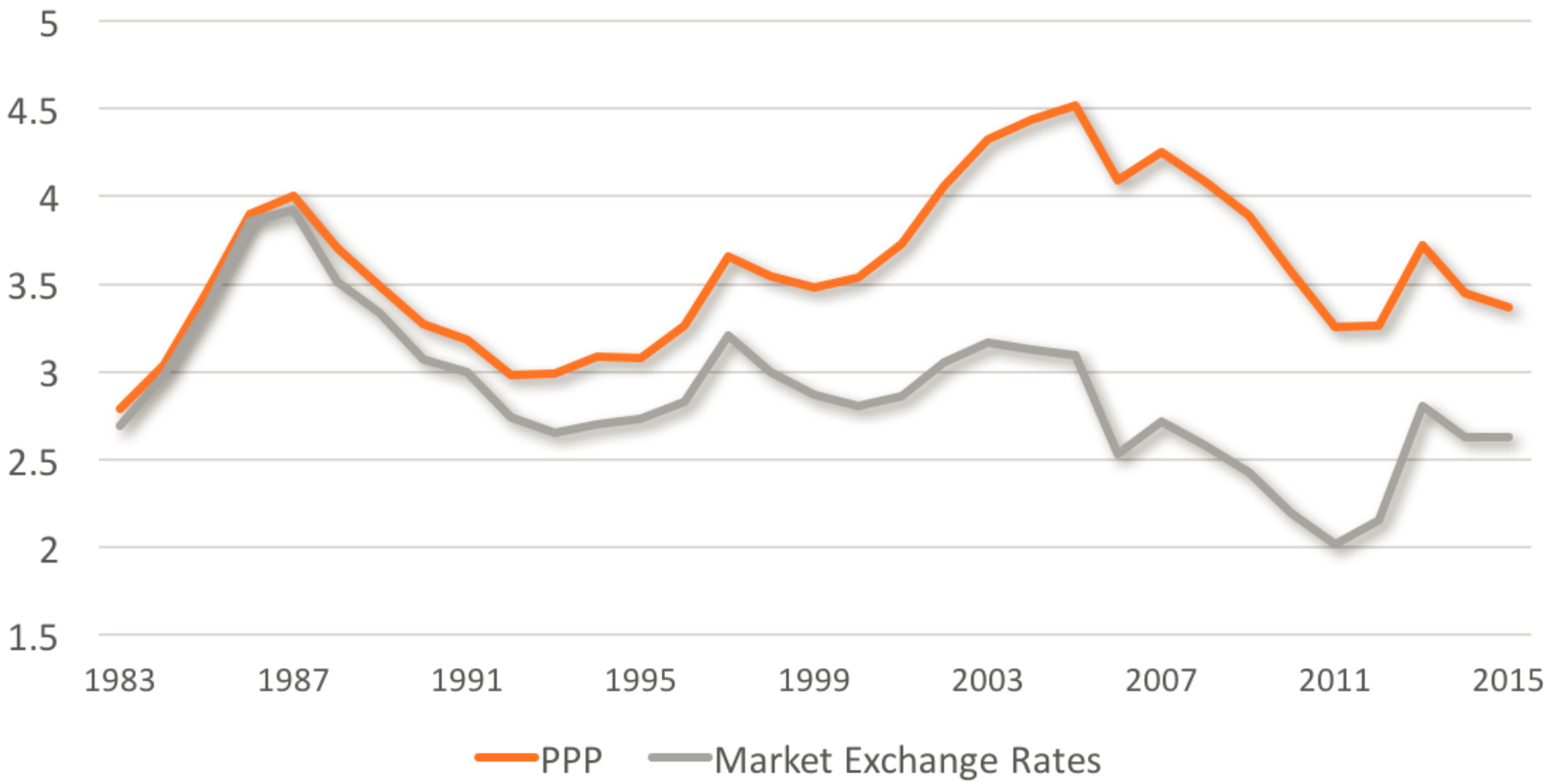
- **How to compute PPP exchange rate?**
  - Need prices of identical baskets
    - Big Mac index, iPhone index, etc.
    - The International Comparison Program (the World Bank)

# PPP and Income Comparison




- The share of rich economies output in the world output is considerably smaller when the PPP conversion is used as opposed to the market exchange rates!

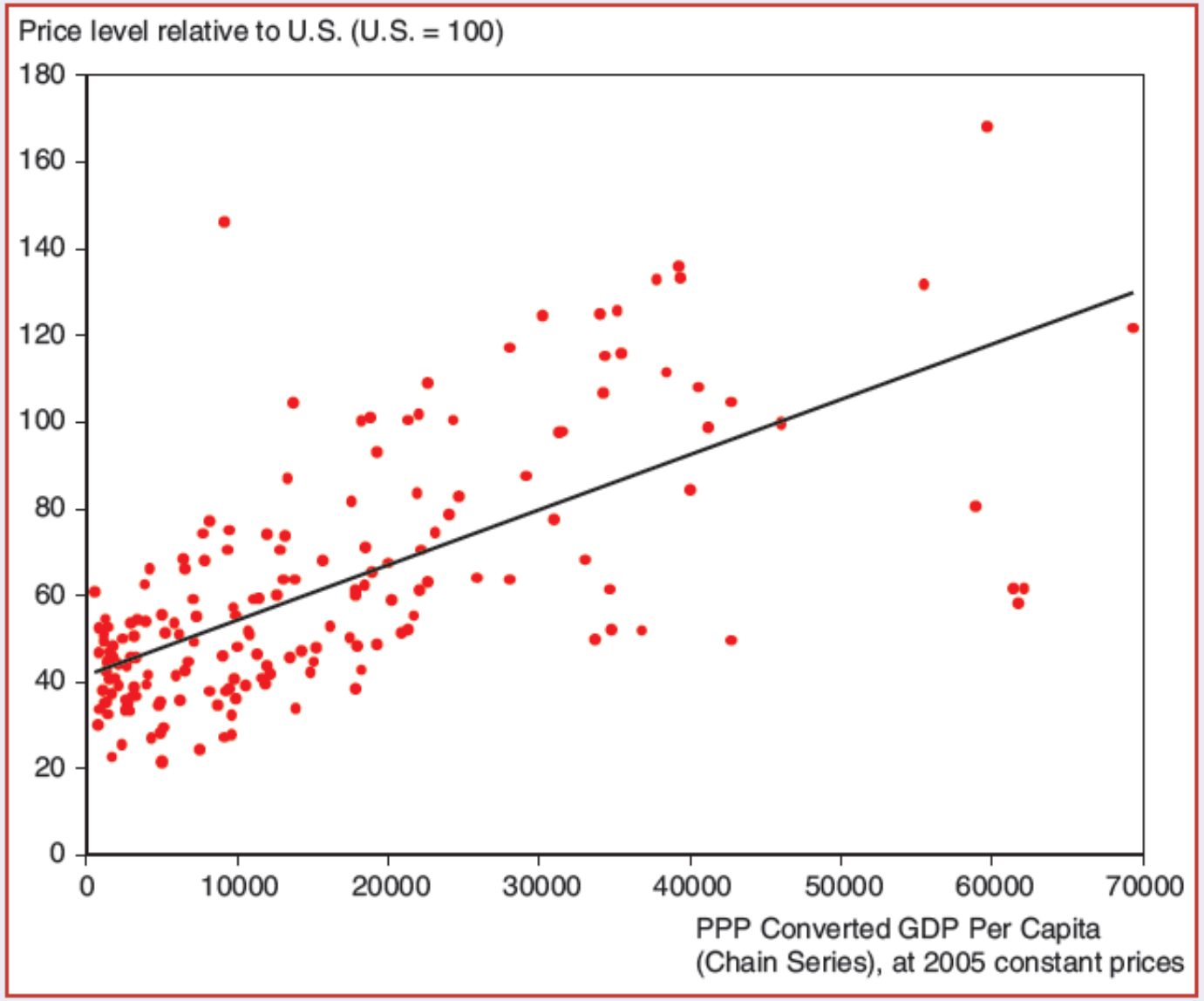
# Exam Question




The figure plots world real GDP annual growth rate. The bottom gray line uses market exchange rate to compute world GDP, while the top orange line uses PPP-adjusted GDP. What can potentially explain the higher growth rate of the PPP adjusted series compared the non-PPP adjusted series?

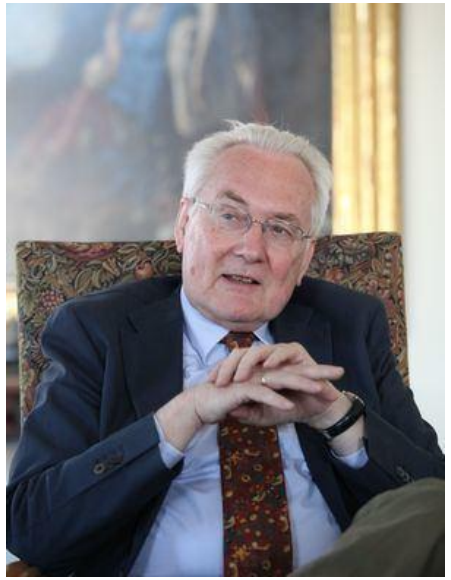
# Price levels and Income

# A Striking Regularity

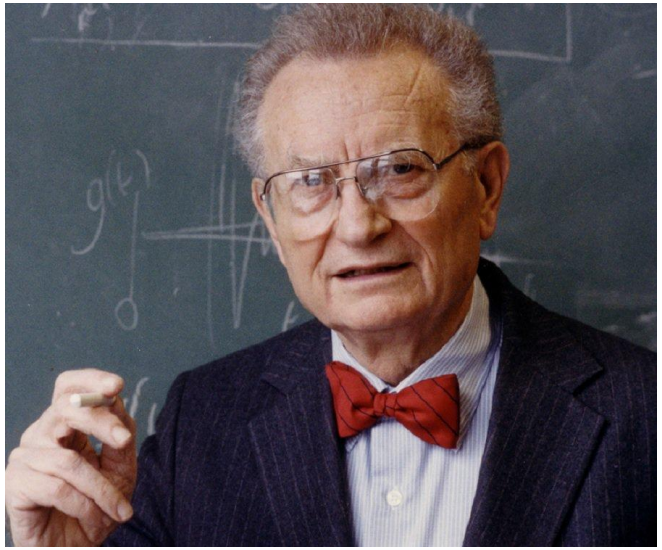



Price levels are lower in poor countries. Why?

# The Balassa-Samuelson Effect



**Béla Balassa (1928-1991)**



**Paul Samuelson (1915-2009)**

- Nobel Prize 1970 (he "*has done more than any other contemporary economist to raise the level of scientific analysis in economic theory*")

# The Balassa-Samuelson Effect

- **Assumptions**
  - Poor countries are less productive in tradable goods than rich countries (cars, computers, etc.)
  - Productivity differences are negligible in the non-traded sector (services: haircuts, nannies, etc.)
- **Intuition**
  - If traded goods satisfy the Law of One Price (LOP), then lower traded goods productivity ⇒ lower wages (in terms of traded goods) ⇒ lower prices of non-traded goods (in terms of traded goods) ⇒ the overall price level is lower (relative to a foreign country)

# The Balassa-Samuelson Effect

## (notes posted online)

### International and Monetary Economics 30054

**The Balassa-Samuelson Effect (Lecture 9)**

**Notation.**

- $i$ - a country index;
- $j$ - a sector index which can be either $N$ (non-traded) or $T$ (traded);
- $Y_i^j$ - output of sector $j$ in country $i$;
- $A_i^j$ - labor productivity of sector $j$ in country $i$;
- $L_i^j$ - employment of sector $j$ in country $i$;
- $P_i^j$ - nominal price of output of sector $j$ in country $i$;
- $P_i$ - aggregate price index in country $i$;
- $W_i$ - nominal wage in country $i$;

**Technology.** Firms produce output according to the following linear production function

$$Y_i^j = A_i^j L_i^j.$$

**Firm's problem.** A (representative) firm in sector $j$ and country $i$ maximizes its static profits

$$\max_{L_i^j} \; P_i^j A_i^j L_i^j - W_i L_i^j.$$

**Equilibrium.** When firms optimize and the markets clear, it must be that marginal revenue equals the marginal cost. Formally,

$$P_i^T A_i^T = P_i^N A_i^N = W_i.$$

1

**Price index in country $i$.** Assume that the price level in country $i$ is weighted (geometric) average of traded and non-traded goods prices:

$$P_i = \left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}.$$

The ratio of price indexes in two countries $i$ and $k$

$$\frac{P_i}{P_k} = \frac{\left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}}{\left(P_k^T\right)^{\alpha} \left(P_k^N\right)^{1-\alpha}} = \frac{P_i^T \left(P_i^N/P_i^T\right)^{1-\alpha}}{P_k^T \left(P_k^N/P_k^T\right)^{1-\alpha}} = \frac{P_i^T \left(A_i^T/A_i^N\right)^{1-\alpha}}{P_k^T \left(A_k^T/A_k^N\right)^{1-\alpha}} = E_{i/k} \frac{\left(A_i^T/A_i^N\right)^{1-\alpha}}{\left(A_k^T/A_k^N\right)^{1-\alpha}}.$$

The last equality assumed that the law of one price holds for traded goods, i.e., $P_i^T/P_k^T = E_{i/k}$. After dividing the last equation by $E_{i/k}$, we obtain

$$\frac{P_i}{P_k E_{i/k}} = \frac{\left(A_i^T/A_i^N\right)^{1-\alpha}}{\left(A_k^T/A_k^N\right)^{1-\alpha}}.$$

Balassa and Samuelson assumed that productivity of non-traded sectors are similar across countries, that is, $A_i^N = A^N$ for all $i$. As a result,

$$\frac{P_i}{P_k E_{i/k}} = \left(\frac{A_i^T}{A_k^T}\right)^{1-\alpha}.$$

The last equation is the main result. It states that the price level in country $i$ relative to the price level in country $k$, after converting it into currency of country $i$, is positively related to the ratio of productivities in traded sectors.

2

# The Balassa-Samuelson Effect

- Price index in country *i*

$$P_i = \left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}$$

- Ratio of price indexes in countries *i* and *k (+algebra)*

$$\frac{P_i}{P_k} = \frac{\left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}}{\left(P_k^T\right)^{\alpha} \left(P_k^N\right)^{1-\alpha}} = \frac{P_i^T \left(P_i^N / P_i^T\right)^{1-\alpha}}{P_k^T \left(P_k^N / P_k^T\right)^{1-\alpha}}$$

# The Balassa-Samuelson Effect

- Technology

$$Y_i^j = A_i^j L_i^j$$

Sector index: T (tradable) or NT (non-tradable)

Country index: e.g., Italy

- Firms

$$\max_{L_i^j} \; P_i^j A_i^j L_i^j - W_i L_i^j$$

Wage does not depend on sector j because workers can switch sectors

- Equilibrium

$$P_i^T A_i^T = P_i^N A_i^N = W_i \longrightarrow \frac{P_i^N}{P_i^T} = \frac{A_i^T}{A_i^N}$$

# The Balassa-Samuelson Effect

- The LOP is assumed to hold for traded goods

$$P_i^T / P_k^T = E_{i/k}$$

- Price indexes again

$$\frac{P_i}{P_k} = \frac{(P_i^T)^{\alpha}(P_i^N)^{1-\alpha}}{(P_k^T)^{\alpha}(P_k^N)^{1-\alpha}} = \frac{P_i^T(P_i^N/P_i^T)^{1-\alpha}}{P_k^T(P_k^N/P_k^T)^{1-\alpha}} = \boxed{\frac{P_i^T(A_i^T/A_i^N)^{1-\alpha}}{P_k^T(A_k^T/A_k^N)^{1-\alpha}} = E_{i/k}\frac{(A_i^T/A_i^N)^{1-\alpha}}{(A_k^T/A_k^N)^{1-\alpha}}}$$

$$\frac{P_i}{P_k E_{i/k}} = \frac{(A_i^T/A_i^N)^{1-\alpha}}{(A_k^T/A_k^N)^{1-\alpha}} \xrightarrow{\boxed{A_i^N = A^N}} \frac{P_i}{P_k E_{i/k}} = \left(\frac{A_i^T}{A_k^T}\right)^{1-\alpha}$$

# The Bhagwati-Kravis-Lipsey Effect

- Rich countries have higher level of capital relative to labor compared to poorer countries

$$\left(\frac{K}{L}\right)^{rich} > \left(\frac{K}{L}\right)^{poor} \Rightarrow$$

- Higher labor productivity (marginal product of labor to be precise)

$$MPL^{rich} > MPL^{poor} \Rightarrow$$

- Higher wages: $w^{rich} > w^{poor} \Rightarrow$
- Because non-tradable goods are more labor intensive, prices of non-tradable goods are higher in richer countries $\Rightarrow$
- Overall price level is higher in richer countries

# Next Time

- KOM, Ch. 17